```
float temperature() {
  float voltage = (analogRead(temperaturePin) / 1024.0) * 5.0;
  float celsius = (voltage - 0.5) * 100.0;
  float fahrenheit = (celsius * 1.8) + 32.0;
  if (degreesF == true) return fahrenheit;
  else return celsius;
}
```

The `temperature()` function is a different data type than what we normally see, so it can return a value that is of the `float` data type. The first line of this function reads the temperature sensor and converts the values from a range of 0–1024 to a voltage from 0 to 5 volts. To obtain the degrees in Celsius we need to offset 0.5v for the TMP36, as described in the device's data sheet, and then multiply this value by 100.0, as shown in Figure 7-2. Once we have the degrees Celsius, we can convert this value to Fahrenheit by multiplying 1.8 and adding 32. The function ends by checking whether or not we wanted degrees in Fahrenheit and returns the appropriate temperature value.

Our next three functions are used to send specific commands to the BlinkM to change its default settings and output the correct color. The specific commands can be found in the BlinkM data sheet available from ThingM's web site. Our first function, `stopScript()`, disables the default startup script by sending it the character "o" as follows:

```
void stopScript(byte address) {
  Wire.beginTransmission(address);
  Wire.write('o');
  Wire.endTransmission();
}
```

Our next function is not entirely necessary, but it sets the BlinkM to its slowest fade speed. Each time we tell the BlinkM to change color, it will fade from its current color to the new one. This is a nice little feature and by slowing it down even more, we help make our device more ambient, slowly changing colors in the background. The `setFadeSpeed()` function will send the BlinkM the character "f" followed by the fade speed with a possible range of 1–255, with 1 being the slowest and 255 being instantaneous. The default speed is 15.

```
void setFadeSpeed(byte address, byte fadespeed) {
  Wire.beginTransmission(address);
  Wire.write('f');
  Wire.write(fadespeed);
  Wire.endTransmission();
}
```

The last function, `fadeToHSB()`, is what makes things so easy for us and is the reason for choosing the BlinkM over the standard RGB LED.

```
void fadeToHSB(byte address, byte hue, byte saturation, byte brightness) {
  Wire.beginTransmission(address);
  Wire.write('h');
  Wire.write(hue);
  Wire.write(saturation);
  Wire.write(brightness);
  Wire.endTransmission();
}
```